case (blank: the method did not run on that case)
C01
C02
C03
C05
C06
C07
C08
C09
C10
C12
C13
C14
Superposition of prisms and lenses
64 2.7e+02 9.3e+02 79 27 1.2e+02 78 2.7e+02 25 69
Random superposition
35 40 39 43 9.5 38 41 14 26
Random mask
4.4 54 15 51
Gerchberg-Saxton
17 17 18 19 8.2 18 24 15 18 11 17
Generalized adaptive additive GS
8.6 13 14 13
Weighted GS
<0.01 1.2 1.1 1.2 0.13 0.76 14 6 0.98 1.1 0.66 0.49
Phase-fixed weighted GS
<0.01 0.38 0.42 <0.01 <0.01 13 6.2 0.022 0.51 0.26
Direct search
<0.01 0.014 <0.01 <0.01 <0.01
Conjugate-gradient minimization
52 41 53 45 63 32 30
Wirtinger gradient descent with Adam
1.3 2.5 1.8 2.5 2
Closed-loop weighted GS
18 20 20 19
Device-aware CG with pixel crosstalk
25
Zernike correction and sensorless modal recovery
0.33
Laguerre-Gaussian vortex traps
1.2 0.13
Hungarian assignment and linear phase interpolation
Multi-plane CNN hologram
36
Volumetric 3D GS
62 16
Device-aware weighted GS
6.5
Flicker-bounded phase-continuous sequence
0.02
Learned warm start for phase-fixed WGS
<0.01 0.38 0.42 <0.01 0.26
Aberration- and pointing-robust hologram
0.59 0.85 0.98 13 5.3
Efficiency-uniformity Pareto front
0.031 0.53 0.66 0.54
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
relative standard deviation of the trap intensities (%)